

Sid Mathur's Thematic ETF Screener – Daily Report

2026-07-25

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Action Center**Run date:** 2026-07-25 **Universe:** 659 ETFs / 26 clusters**Full PC1:** FLAT **Clusters long:** 3/26 **ETFs long:** 155/659 (24%)**Today's events (cluster level)**

spread_buy = 5dma – 50dma; spread_sell = 5dma – 20dma (price units). A BUY fired today iff spread_buy crossed above 0; a SELL iff spread_sell crossed below the cushioned threshold.

No new cluster crossovers today.

Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p_h5 is the 5-day probability for context. **spread_buy = 5dma – 50dma, spread_sell = 5dma – 20dma** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

Imminent BUY (top 5 by h=1 NSS, filter-approved)

cluster	theme	spread_buy	p_h1	p_h5	nss_h1	nss_h5
cluster_9	U.S. Dividend Equity	3.523	1	1.00	1.000	1.00
cluster_15	Real Estate	3.290	1	1.00	1.000	1.00
cluster_14	Financials	5.456	1	1.00	0.989	0.78
cluster_11	Healthcare	4.475	1	1.00	0.014	0.07
cluster_2	U.S. Mid Cap Equity	1.896	1	0.97	0.013	0.03

Imminent SELL (top 1 by h=1 NSS)

cluster	theme	spread_sell	p_h1	p_h5	nss_h1	nss_h5
cluster_1	Developed ex-US Equity	0.063	0.32	0.27	-0.193	0.09

Full Sample PC1

Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

Replication basket: VOO basket cor: 0.94 sticky

Current state: FLAT

5dma: 141.552 20dma: 142.249 50dma: 141.770

Spread vs 50dma: -0.218 Spread vs 20dma: -0.697

P(buy crossover within 5 days): 8.0% P(sell): 95.6%

Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

Active positions (long clusters)

ticker	cluster	theme	nss	days	return	pos_wt	cor
PDBC	cluster_19	Broad Commodities	1.000	2	1.2%	22.45%	0.98
XLP	cluster_18	Consumer Staples	0.482	19	-0.2%	11.76%	0.99
XLU	cluster_18	Consumer Staples	0.482	19	-0.2%	7.26%	0.99
VEA	cluster_1	Developed ex-US Equity	0.089	11	0.1%	18.13%	0.99

Watchlist (imminent-buy clusters, p > 80%, not yet long)

ticker	cluster	theme	p_h5	nss	weight	cor
XLF	cluster_14	Financials	1.00	0.783	1.095	0.96
VHT	cluster_11	Healthcare	1.00	0.068	1.092	0.93
VNQ	cluster_15	Real Estate	1.00	0.997	0.970	0.99
SCHD	cluster_9	U.S. Dividend Equity	1.00	0.999	0.893	0.96
SQQQ	cluster_26	Leveraged / Inverse	1.00	0.996	0.506	1.00
UUP	cluster_26	Leveraged / Inverse	1.00	0.996	0.495	1.00
IJH	cluster_2	U.S. Mid Cap Equity	0.97	0.025	1.003	0.99
VTI	cluster_3	U.S. Large Cap Value	0.93	-0.030	1.093	0.97

Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

	id	name	category	n	signal	change	days	return	nss	dNSS
19	cluster_19	Broad Commodities	Broad Commodities	11	LONG		2	1.2%	1.000	+0.000
9	cluster_9	U.S. Dividend Equity	U.S. Dividend Equ...	25	FLAT				0.999	-0.001
15	cluster_15	Real Estate	Real Estate	14	FLAT				0.997	-0.000
26	cluster_26	Leveraged / Inverse	Leveraged / Inver...	2	FLAT	shifted			0.996	+0.004
14	cluster_14	Financials	Financials	14	FLAT				0.783	-0.114
18	cluster_18	Consumer Staples	Consumer Staples	11	LONG		19	-0.2%	0.482	-0.098
1	cluster_1	Developed ex-US Equity	Developed ex-US E...	85	LONG		11	0.1%	0.089	+0.096
12	cluster_12	Materials	Materials	17	FLAT				0.089	+0.083
11	cluster_11	Healthcare	Healthcare	17	FLAT				0.068	-0.071
2	cluster_2	U.S. Mid Cap Equity	U.S. Mid Cap Equi...	72	FLAT				0.025	-0.110
3	cluster_3	U.S. Large Cap Value	U.S. Large Cap Va...	62	FLAT				-0.030	+0.046
13	cluster_13	China Equity	China Equity	15	FLAT				-0.075	+0.105
17	cluster_17	Gold	Gold	12	FLAT				-0.321	+0.293
24	cluster_24	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	9	FLAT				-0.460	-0.159
5	cluster_5	U.S. Large Cap Equity	U.S. Large Cap Eq...	47	FLAT				-0.698	-0.134
20	cluster_20	India Equity	India Equity	10	FLAT				-0.849	-0.359
25	cluster_25	International Fixed Income	International Fix...	6	FLAT				-0.870	-0.201
22	cluster_22	Japan Equity	Japan Equity	9	FLAT				-0.937	+0.055
21	cluster_21	Inflation-Protected Fixed I...	Inflation-Protect...	10	FLAT				-0.959	-0.037
8	cluster_8	Emerging Markets Equity	Emerging Markets ...	25	FLAT				-0.983	+0.012
23	cluster_23	Bank Loan / Floating Rate	Bank Loan / Float...	9	FLAT				-0.985	-0.073
6	cluster_6	U.S. Large Cap Growth	U.S. Large Cap Gr...	44	FLAT				-0.997	+0.001
7	cluster_7	U.S. Extended Market Equity	U.S. Extended Mar...	41	FLAT				-0.999	-0.001
4	cluster_4	Aggregate Fixed Income	Aggregate Fixed I...	58	FLAT				-1.000	-0.001
10	cluster_10	High Yield Bond	High Yield Bond	21	FLAT				-1.000	-0.000
16	cluster_16	Municipal Bond	Municipal Bond	13	FLAT				-1.000	-0.000

Cluster Details

Cluster 1 – Developed ex-US Equity

Members: 85 **Var explained:** 0.814 **Signal:** LONG since 2026-07-09 (11 days, 0.1%)

Signal logic:

- Latest buy_cross: 2026-07-09 – filter **PASS** (252d at date = +17.9%)
- Latest sell_cross: 2026-06-24
- Current 252d return: +15.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-07-09 > last sell at 2026-06-24)



Category breakdown

category	n_members	abs_weight
International Equity	21	0.2625
Developed ex-US Equity	9	0.1153
International Dividend Equity	6	0.0730
Global ex-US Equity	5	0.0641
Europe Equity	4	0.0507
Alternatives	4	0.0454
Eurozone Equity	3	0.0364
Asia-Pacific Equity	2	0.0238

Real Estate	2	0.0220
Global Equity	2	0.0216
Emerging Markets Fixed Income	2	0.0191
Currency	2	0.0177
Commodity	1	0.0130
Global ex-US Small Cap Equity	1	0.0126
Germany Equity	1	0.0121
France Equity	1	0.0121
United Kingdom Equity	1	0.0120
Italy Equity	1	0.0119
Europe Financials	1	0.0118
Sweden Equity	1	0.0117
Australia Equity	1	0.0117
Netherlands Equity	1	0.0116
Spain Equity	1	0.0113
Canada Equity	1	0.0112
Switzerland Equity	1	0.0111
Austria Equity	1	0.0110
Belgium Equity	1	0.0110
Singapore Equity	1	0.0103
Poland Equity	1	0.0102
South Africa Equity	1	0.0098
Latin America Equity	1	0.0090
New Zealand Equity	1	0.0089
Mexico Equity	1	0.0088
Chile Equity	1	0.0078
Brazil Equity	1	0.0074

Replication basket – VEA (cor=0.99) basket lead at lag 1: -0.012

ticker	rep_wt	name	category	etf_signal
VEA	0.906	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
3	EFA	0.0130	iShares MSCI EAFE ETF	Developed ex-US Equity	LONG
4	VEA	0.0130	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT
51	GSIE	0.0130	Goldman Sachs ActiveBeta International ...	Commodity	LONG
54	ESGD	0.0130	iShares ESG Aware MSCI EAFE ETF	International Equity	LONG
65	IEFA	0.0130	iShares Core MSCI EAFE ETF	Developed ex-US Equity	LONG

67	SPDW	0.0130	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	FLAT
23	DWM	0.0129	WisdomTree International Equity Fund	International Dividend E...	FLAT
36	TLTD	0.0129	FlexShares Morningstar Developed Market...	International Equity	FLAT
43	INTF	0.0129	iShares International Equity Factor ETF	International Equity	LONG
66	SCHF	0.0129	Schwab International Equity ETF	Developed ex-US Equity	FLAT
30	PXF	0.0128	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	FLAT
40	ACWX	0.0128	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	FLAT
41	CWI	0.0128	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	FLAT
44	IQLT	0.0128	iShares MSCI Intl Quality Factor ETF	International Equity	LONG
46	QEFA	0.0128	State Street SPDR MSCI EAFE StrategicFa...	International Equity	LONG
49	VYMI	0.0128	Vanguard International High Dividend Yi...	International Dividend E...	LONG
50	FNDF	0.0128	Schwab Fundamental International Equity...	International Equity	FLAT
60	IEUR	0.0128	iShares Core MSCI Europe ETF	Europe Equity	LONG
61	IXUS	0.0128	iShares Core MSCI Total International S...	Global ex-US Equity	FLAT
62	VEU	0.0128	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	FLAT
63	VGK	0.0128	Vanguard FTSEEuropean ETF	Europe Equity	LONG
64	VXUS	0.0128	Vanguard Total International Stock ETF	Global ex-US Equity	FLAT
7	IEV	0.0127	iShares Europe ETF	Europe Equity	LONG
34	EFV	0.0127	iShares MSCI EAFE Value ETF	Developed ex-US Equity	LONG
35	FNDC	0.0127	Schwab Fundamental International Small ...	International Equity	FLAT
37	JPIN	0.0127	JPMorgan Diversified Return Internation...	International Equity	FLAT
48	RODM	0.0127	Hartford Multifactor Developed Markets ...	International Equity	LONG
58	SCHC	0.0127	Schwab International Small-Cap Equity E...	Developed ex-US Equity	FLAT
26	IQDF	0.0126	FlexShares International Quality Divide...	International Equity	LONG
33	SCZ	0.0126	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	FLAT
59	VSS	0.0126	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	FLAT
9	DLS	0.0125	WisdomTree International SmallCap Fund	International Equity	FLAT
21	EZU	0.0125	iShares MSCI Eurozone ETF	Eurozone Equity	FLAT
45	IVLU	0.0125	iShares MSCI Intl Value Factor ETF	International Equity	LONG
5	FEP	0.0124	First Trust Europe AlphaDEX Fund	Europe Equity	FLAT
31	EFG	0.0124	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	FLAT
47	FEZ	0.0124	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	FLAT
56	VIGI	0.0124	Vanguard International Dividend Appreci...	International Dividend E...	LONG
76	HFXI	0.0124	NYLI FTSE International Equity Currency...	International Equity	FLAT
32	GWX	0.0123	State Street SPDR S&P International Sma...	International Equity	FLAT
42	IMTM	0.0123	iShares MSCI Intl Momentum Factor ETF	International Equity	FLAT
1	EWG	0.0121	iShares MSCI Germany Index Fund	Germany Equity	FLAT
6	VIDI	0.0121	Vident International Equity Strategy ETF	International Equity	FLAT
17	EWQ	0.0121	iShares MSCI France Index Fund	France Equity	FLAT
22	IDV	0.0121	iShares International Select Dividend E...	International Dividend E...	FLAT
82	IPKW	0.0121	Invesco International BuyBack Achievers...	International Equity	LONG
19	EWU	0.0120	iShares MSCI United Kingdom ETF	United Kingdom Equity	LONG
55	HDEF	0.0120	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	LONG
24	EPP	0.0119	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	LONG
52	IDLV	0.0119	Invesco S&P International Developed Low...	International Equity	LONG

84	BBAX	0.0119	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	LONG
85	EWI	0.0119	iShares MSCI Italy ETF	Italy Equity	FLAT
25	EUFN	0.0118	iShares MSCI Europe Financials ETF	Europe Financials	LONG
14	EWA	0.0117	iShares MSCI Australia Index Fund	Australia Equity	LONG
53	EFAV	0.0117	iShares MSCI EAFE Min Vol Factor ETF	International Equity	LONG
75	EWD	0.0117	iShares MSCI Sweden ETF	Sweden Equity	FLAT
29	DWX	0.0116	State Street SPDR S&P International Div...	International Dividend E...	LONG
38	DBEF	0.0116	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
80	EWN	0.0116	iShares MSCI Netherlands Index Fund	Netherlands Equity	FLAT
13	HEZU	0.0115	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	FLAT
39	HEFA	0.0115	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
12	PID	0.0113	Invesco International Dividend Achiever...	International Dividend E...	LONG
16	EWP	0.0113	iShares MSCI Spain ETF	Spain Equity	FLAT
11	IHDG	0.0112	WisdomTree International Hedged Quality...	Alternatives	FLAT
20	EWK	0.0112	iShares MSCI Canada Index Fund	Canada Equity	LONG
57	VNQI	0.0112	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	LONG
10	HEDJ	0.0111	WisdomTree Europe Hedged Equity Fund	Alternatives	FLAT
15	EWL	0.0111	iShares MSCI Switzerland ETF	Switzerland Equity	FLAT
27	NFRA	0.0111	FlexShares STOXX Global Broad Infrastru...	Global Equity	LONG
79	EWO	0.0110	iShares MSCI Austria ETF	Austria Equity	LONG
83	EWK	0.0110	iShares MSCI Belgium ETF	Belgium Equity	FLAT
8	RWX	0.0108	State Street SPDR Dow Jones Internation...	Real Estate	LONG
28	IGF	0.0105	iShares Global Infrastructure ETF	Global Equity	LONG
18	EWS	0.0103	iShares MSCI Singapore ETF	Singapore Equity	LONG
74	EPOL	0.0102	iShares MSCI Poland ETF	Poland Equity	LONG
78	EZA	0.0098	iShares MSCI South Africa Index Fund	South Africa Equity	FLAT
68	EBND	0.0095	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	FLAT
69	EMLC	0.0095	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	FLAT
73	FXA	0.0095	Invesco CurrencyShares Australian Dolla...	Currency	FLAT
72	ILF	0.0090	iShares Latin America 40 ETF	Latin America Equity	LONG
77	ENZL	0.0089	iShares MSCI New Zealand ETF	New Zealand Equity	LONG
70	EWX	0.0088	iShares MSCI Mexico ETF	Mexico Equity	FLAT
81	FXC	0.0082	Invesco CurrencyShares Canadian Dollar ...	Currency	FLAT
71	ECH	0.0078	iShares MSCI Chile ETF	Chile Equity	FLAT
2	EWZ	0.0074	iShares MSCI Brazil ETF	Brazil Equity	LONG

Cluster 2 – U.S. Mid Cap Equity

Members: 72 **Var explained:** 0.885 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +38.9%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +19.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
U.S. Small Cap Equity	10	0.1448
Consumer Discretionary	9	0.1141
U.S. Mid Cap Equity	6	0.0870
U.S. Small Cap Value	6	0.0865
Industrials	4	0.0504
U.S. Mid Cap Value	3	0.0436
U.S. Small Cap Growth	3	0.0433
U.S. Mid Cap Growth	3	0.0422
Leveraged / Inverse	2	0.0278
Dividend Equity	2	0.0276
U.S. Small Cap Low Volatility	2	0.0263
U.S. Small Cap Fundamental Equity	1	0.0147
U.S. Small Cap ESG Equity	1	0.0146
Unknown	1	0.0146
U.S. Mid/Small Cap Equity	1	0.0146
Mid Cap Value Equities	1	0.0146

U.S. Small/Mid Cap Fundamental Equity	1	0.0145
U.S. Small Cap Multifactor Equity	1	0.0145
Small Cap Value Equities	1	0.0143
U.S. Small/Mid Cap Equity	1	0.0143
U.S. Mid Cap Dividend Equity	1	0.0142
U.S. ESG Equity	1	0.0142
Value Equity	1	0.0141
U.S. Small Cap Dividend Equity	1	0.0140
U.S. Mid Cap Quality Equity	1	0.0140
U.S. Equity	1	0.0140
U.S. Small Cap Industrials	1	0.0137
U.S. Small Cap Momentum Equity	1	0.0137
U.S. Micro Cap Equity	1	0.0134
U.S. Small Cap Consumer Discretionary	1	0.0133
U.S. Dividend Equity	1	0.0132
U.S. Mid Cap Low Volatility	1	0.0126
Real Estate	1	0.0114

Replication basket – IJH (cor=0.99) basket lead at lag 1: -0.011

ticker	rep_wt	name	category	etf_signal
IJH	1.003	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
46	FNDA	0.0147	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	FLAT
12	FNX	0.0146	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	FLAT
15	NUSC	0.0146	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	FLAT
18	MDY	0.0146	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	FLAT
25	IVOO	0.0146	Vanguard S&P Mid-Cap 400 ETF		FLAT
26	VI00	0.0146	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	FLAT
28	IJR	0.0146	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	FLAT
29	IJH	0.0146	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
30	SPMD	0.0146	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	FLAT
31	SPSM	0.0146	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	FLAT
32	VB	0.0146	Vanguard Small-Cap ETF	U.S. Small Cap Equity	FLAT
43	EZM	0.0146	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	FLAT
58	VBR	0.0146	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	FLAT
13	FYX	0.0145	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	FLAT
16	PRFZ	0.0145	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	FLAT

17	SMLF	0.0145	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	FLAT
19	IJT	0.0145	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	FLAT
33	SCHA	0.0145	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	FLAT
51	IJJ	0.0145	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
53	IVOV	0.0145	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
54	MDYV	0.0145	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	FLAT
5	IWM	0.0144	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	FLAT
9	JHMM	0.0144	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	FLAT
11	SCHM	0.0144	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	FLAT
21	VIOG	0.0144	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	FLAT
24	SLYG	0.0144	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	FLAT
27	VTWO	0.0144	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	FLAT
49	IWN	0.0144	iShares Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
52	IJS	0.0144	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
57	VTWV	0.0144	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
64	RWK	0.0144	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	FLAT
42	EES	0.0143	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	FLAT
44	PSC	0.0143	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	FLAT
55	SLYV	0.0143	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	FLAT
56	VIOV	0.0143	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	FLAT
60	RWJ	0.0143	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	FLAT
71	SDVY	0.0143	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	LONG
72	ISMD	0.0143	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
41	DON	0.0142	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	FLAT
59	ETHO	0.0142	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	FLAT
20	IJK	0.0141	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	FLAT
22	IVOG	0.0141	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	FLAT
23	MDYG	0.0141	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	FLAT
70	FYT	0.0141	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	FLAT
40	DES	0.0140	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	FLAT
62	XMHQ	0.0140	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	FLAT
63	RFV	0.0140	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	FLAT
65	OMFS	0.0140	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	FLAT
34	FXD	0.0138	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	FLAT
66	RZV	0.0138	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	FLAT
61	PSCI	0.0137	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	FLAT
69	XSMO	0.0137	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	FLAT
38	RSPD	0.0135	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	FLAT
14	IWC	0.0134	iShares Microcap ETF	U.S. Micro Cap Equity	FLAT
50	SMMV	0.0134	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	LONG
10	REGL	0.0133	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	LONG
36	FTXR	0.0133	First Trust Nasdaq Transportation ETF	Industrials	LONG
68	PSCD	0.0133	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	FLAT
39	SMDV	0.0132	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	FLAT
45	XTN	0.0130	State Street SPDR S&P Transportation ETF	Industrials	FLAT

48	XSLV	0.0130	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	FLAT
2	XRT	0.0128	State Street SPDR S&P Retail ETF	Consumer Discretionary	LONG
6	IYC	0.0128	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	FLAT
7	VCR	0.0126	Vanguard Consumer Discretion ETF	Consumer Discretionary	FLAT
8	FDIS	0.0126	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	FLAT
37	IYT	0.0126	iShares U.S. Transportation ETF	Industrials	FLAT
47	XMLV	0.0126	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	LONG
35	PEJ	0.0125	Invesco Leisure and Entertainment ETF	Consumer Discretionary	FLAT
1	XLY	0.0121	State Street Consumer Discretionary Sel...	Consumer Discretionary	FLAT
3	JETS	0.0115	U.S. Global Jets ETF	Industrials	FLAT
4	REM	0.0114	iShares Mortgage Real Estate ETF	Real Estate	FLAT
67	RTH	0.0113	VanEck Retail ETF	Consumer Discretionary	LONG

Cluster 3 – U.S. Large Cap Value

Members: 62 **Var explained:** 0.878 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +27.9%)
- Latest sell_cross: 2026-07-22
- Current 252d return: +17.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-22 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
U.S. Large Cap Value	10	0.1684
Industrials	10	0.1521
U.S. Dividend Equity	8	0.1325
Materials	4	0.0613
U.S. Mid Cap Equity	3	0.0502
U.S. Multifactor Equity	3	0.0499
U.S. Value Factor Equity	3	0.0490
Utilities	3	0.0457
U.S. Fundamental Equity	2	0.0339
U.S. Equal Weight Equity	2	0.0338
U.S. Mid Cap Value	2	0.0338
U.S. Equity	2	0.0334
Consumer Discretionary	2	0.0264
Large Cap Blend Equities	1	0.0169
U.S. Size Factor Equity	1	0.0168
U.S. Revenue Weighted Equity	1	0.0167
Value Equity	1	0.0166
U.S. Mega Cap Value	1	0.0165
U.S. Buyback Equity	1	0.0164
U.S. Large Cap Equity	1	0.0163
Communication Services	1	0.0134

Replication basket – VTV (cor=0.97) basket lead at lag 1: -0.002

ticker	rep_wt	name	category	etf_signal
VTV	1.093	Vanguard Value ETF	U.S. Large Cap Value	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
3	RSP	0.0171	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
14	PRF	0.0170	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
17	CFA	0.0170	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	FLAT
25	IWD	0.0170	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
35	VONV	0.0170	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
41	SCHV	0.0170	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	FLAT
50	IWS	0.0170	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
6	FEX	0.0169	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
21	FNDX	0.0169	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
28	JPUS	0.0169	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	FLAT
22	EQAL	0.0168	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	FLAT
26	IWR	0.0168	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
27	IVE	0.0168	iShares S&P 500 Value ETF	U.S. Large Cap Value	FLAT
29	SIZE	0.0168	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	FLAT
33	VOOV	0.0168	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	FLAT
38	ILCV	0.0168	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	FLAT
39	IMCB	0.0168	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
40	IUSV	0.0168	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	FLAT
44	VTV	0.0168	Vanguard Value ETF	U.S. Large Cap Value	LONG
51	VOE	0.0168	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
61	JPME	0.0168	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	FLAT
15	RWL	0.0167	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
19	NULV	0.0167	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	FLAT
42	SPYV	0.0167	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	FLAT
43	VO	0.0167	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
54	DTD	0.0167	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
56	DGRO	0.0167	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
62	JVAL	0.0167	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	FLAT
11	IYJ	0.0166	iShares U.S. Industrials ETF	Industrials	FLAT
23	IWX	0.0166	iShares Russell Top 200 Value ETF	Value Equity	LONG
24	ONEV	0.0166	State Street SPDR Russell 1000 Low Vola...	U.S. Equity	FLAT
37	VYM	0.0166	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	FLAT
52	PFM	0.0166	Invesco Dividend Achievers ETF	U.S. Dividend Equity	FLAT
53	DLN	0.0166	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG

57	VIG	0.0166	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
16	AIVL	0.0165	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
36	MGV	0.0165	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	FLAT
4	FXR	0.0164	First Trust Industrials AlphaDEX	Industrials	FLAT
5	PKW	0.0164	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	LONG
10	RDVY	0.0164	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	FLAT
20	FDVV	0.0164	Fidelity High Dividend ETF	U.S. Dividend Equity	FLAT
32	VIS	0.0163	Vanguard Industrials ETF	Industrials	LONG
34	FIDU	0.0163	Fidelity MSCI Industrials Index ETF	Industrials	LONG
55	DIA	0.0163	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	FLAT
1	XLI	0.0162	State Street Industrial Select Sector S...	Industrials	FLAT
18	CFO	0.0160	VictoryShares US 500 Enhanced Volatilit...	U.S. Multifactor Equity	FLAT
58	PAVE	0.0159	Global X U.S. Infrastructure Developmen...	Industrials	FLAT
30	VLUE	0.0158	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	FLAT
31	VAW	0.0157	Vanguard Materials ETF	Materials	FLAT
9	FIW	0.0156	First Trust Water ETF	Utilities	FLAT
2	XLB	0.0155	State Street Materials Select Sector SP...	Materials	FLAT
7	PHO	0.0155	Invesco Water Resources ETF	Utilities	FLAT
13	RSPM	0.0153	Invesco S&P 500 Equal Weight Materials ...	Materials	LONG
12	IYM	0.0149	iShares U.S. Basic Materials ETF	Materials	FLAT
8	CGW	0.0146	Invesco S&P Global Water Index ETF	Utilities	FLAT
60	PKB	0.0145	Invesco Building & Construction ETF	Industrials	FLAT
45	PPA	0.0138	Invesco Aerospace & Defense ETF	Industrials	FLAT
49	XHB	0.0138	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	FLAT
59	IYZ	0.0134	iShares U.S. Telecommunications ETF	Communication Services	FLAT
48	XAR	0.0133	State Street SPDR S&P Aerospace & Defen...	Industrials	FLAT
46	ITA	0.0129	iShares U.S. Aerospace & Defense ETF	Industrials	LONG
47	ITB	0.0126	iShares U.S. Home Construction ETF	Consumer Discretionary	FLAT

Cluster 4 – Aggregate Fixed Income

Members: 58 **Var explained:** 0.832 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +4.5%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +2.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-17)



Category breakdown

category	n_members	abs_weight
Fixed Income	19	0.3284
Investment Grade Corporate Bond	13	0.2302
Aggregate Fixed Income	6	0.1120
Mortgage-Backed Fixed Income	6	0.0982
Active Fixed Income	5	0.0891
Cash / Ultra-Short Fixed Income	3	0.0471
International Fixed Income	2	0.0313
Target Maturity Bond	2	0.0306
Agency Fixed Income	1	0.0166
Taxable Municipal Bond	1	0.0165

Replication basket – BND (cor=0.99) basket lead at lag 1: 0.007

ticker	rep_wt	name	category	etf_signal
BND	1.007	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	BIV	0.0188	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
26	AGG	0.0188	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	FLAT
28	BND	0.0188	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT
30	SCHZ	0.0188	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	FLAT
20	IUSB	0.0187	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	FLAT
31	SPAB	0.0187	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	FLAT
2	IEF	0.0185	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
17	IGIB	0.0184	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	FLAT
24	VCIT	0.0184	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	FLAT
7	FBND	0.0183	Fidelity Total Bond ETF	Active Fixed Income	FLAT
43	SPTI	0.0183	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
46	SCHR	0.0183	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
47	VGIT	0.0183	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
13	AGGY	0.0182	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	FLAT
15	SPIB	0.0182	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	FLAT
6	GTO	0.0181	Invesco Total Return Bond ETF	Active Fixed Income	FLAT
11	GVI	0.0181	iShares Intermediate Government/Credit ...	Fixed Income	FLAT
40	IEI	0.0181	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
50	MBB	0.0181	iShares MBS ETF	Mortgage-Backed Fixed In...	FLAT
12	QLTA	0.0180	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	FLAT
27	USIG	0.0180	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	FLAT
51	VMBS	0.0180	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	FLAT
52	SPMB	0.0180	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	FLAT
3	LQD	0.0179	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	FLAT
5	BOND	0.0179	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	FLAT
19	ISTB	0.0178	iShares Core 1-5 Year USD Bond ETF	Fixed Income	FLAT
8	FLCO	0.0177	Franklin Investment Grade Corporate ETF	Active Fixed Income	FLAT
10	CORP	0.0177	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	FLAT
22	BLV	0.0176	Vanguard Long-Term Bond ETF	Fixed Income	FLAT
41	TLH	0.0176	iShares 10-20 Year Treasury Bond ETF	Fixed Income	FLAT
45	GOVT	0.0176	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
35	IGSB	0.0175	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
37	VCSH	0.0175	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
23	BSV	0.0174	Vanguard Short-Term Bond ETF	Fixed Income	FLAT
49	GNMA	0.0174	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	FLAT
18	IGLB	0.0172	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	FLAT
16	SPLB	0.0171	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	FLAT
25	VCLT	0.0171	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
36	SLQD	0.0171	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
44	SPTL	0.0171	State Street SPDR Portfolio Long Term T...	Fixed Income	FLAT
48	VGLT	0.0171	Vanguard Long-Term Treasury ETF	Fixed Income	FLAT
29	TOTL	0.0170	State Street DoubleLine Total Return Ta...	Active Fixed Income	FLAT
1	TLT	0.0169	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT

39	AGZ	0.0166	iShares Agency Bond ETF	Agency Fixed Income	FLAT
9	BAB	0.0165	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	FLAT
42	EDV	0.0163	Vanguard Extended Duration Treasury ETF	Fixed Income	FLAT
34	SPSB	0.0162	State Street SPDR Portfolio Short Term ...	Fixed Income	FLAT
53	SHY	0.0162	iShares 1-3 Year Treasury Bond ETF	Fixed Income	FLAT
56	VGSH	0.0159	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
57	BNDX	0.0158	Vanguard Total International Bond ETF	International Fixed Inco...	FLAT
55	SCHO	0.0156	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
58	IAGG	0.0156	iShares International Aggregate Bond Fu...	International Fixed Inco...	FLAT
33	BSCQ	0.0155	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	FLAT
54	SPTS	0.0155	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	FLAT
14	IBDR	0.0151	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	LONG
38	CMBS	0.0135	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	FLAT
4	LMBS	0.0133	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	FLAT
32	LDUR	0.0121	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	FLAT

Cluster 5 – U.S. Large Cap Equity

Members: 47 **Var explained:** 0.948 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +25.6%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +15.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
U.S. Large Cap Equity	9	0.1959
U.S. Total Market Equity	7	0.1525
Global Equity	7	0.1495
Asset Allocation	4	0.0792
U.S. Mega Cap Equity	3	0.0640
U.S. Multifactor Equity	3	0.0634
U.S. Quality / Factor Equity	3	0.0633
U.S. ESG Equity	2	0.0434
U.S. Equity	2	0.0434
Commodity	1	0.0217
Large Cap Value Equities	1	0.0216
Large Cap Blend Equities	1	0.0215
Dividend Equity	1	0.0213
U.S. Dividend Equity	1	0.0209
U.S. Moat Equity	1	0.0197
Alternatives	1	0.0186

Replication basket – VOO (cor=1.00) basket lead at lag 1: -0.010

ticker	rep_wt	name	category	etf_signal
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V00 0.936 Vanguard S&P 500 ETF U.S. Large Cap Equity FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	SPY	0.0218	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	FLAT
21	IWV	0.0218	iShares Russell 3000 Fund	U.S. Total Market Equity	FLAT
22	IYY	0.0218	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	FLAT
27	ESGU	0.0218	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	FLAT
28	IWB	0.0218	iShares Russell 1000 ETF	U.S. Large Cap Equity	FLAT
36	VONE	0.0218	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	FLAT
38	VV	0.0218	Vanguard Large-Cap ETF	U.S. Large Cap Equity	FLAT
39	ILCB	0.0218	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	FLAT
40	ITOT	0.0218	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	FLAT
41	IVV	0.0218	iShares Core S&P 500 ETF	U.S. Large Cap Equity	FLAT
42	SCHB	0.0218	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	FLAT
43	SCHX	0.0218	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	FLAT
44	SPTM	0.0218	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	FLAT
45	V00	0.0218	Vanguard S&P 500 ETF	U.S. Large Cap Equity	FLAT
46	VTI	0.0218	Vanguard Total Stock Market ETF	U.S. Total Market Equity	FLAT
8	URTH	0.0217	iShares MSCI World ETF	Global Equity	LONG
20	SUSA	0.0217	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	FLAT
25	SPYX	0.0217	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	FLAT
33	VTHR	0.0217	Vanguard Russell 3000 ETF	U.S. Total Market Equity	FLAT
34	GSLC	0.0217	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	FLAT
17	JHML	0.0216	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	FLAT
23	LRGF	0.0216	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	FLAT
29	IWL	0.0216	iShares Russell Top 200 ETF	U.S. Equity	FLAT
35	EPS	0.0216	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	FLAT
37	MGC	0.0216	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	FLAT
3	ACWI	0.0215	iShares MSCI ACWI ETF	Global Equity	FLAT
11	VT	0.0215	Vanguard Total World Stock Index ETF	Global Equity	FLAT
13	TILT	0.0215	FlexShares Mornigstar US Market Factors...	U.S. Multifactor Equity	FLAT
19	DSI	0.0215	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	FLAT
9	CRBN	0.0214	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	FLAT
10	SPGM	0.0214	State Street SPDR Portfolio MSCI Global...	Global Equity	FLAT
24	OEF	0.0214	iShares S&P 100 Fund	U.S. Mega Cap Equity	LONG
30	QUAL	0.0214	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	FLAT
4	AOA	0.0213	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	FLAT
15	QDF	0.0213	FlexShares Quality Dividend Index Fund	Dividend Equity	FLAT
14	IOO	0.0211	iShares Global 100 ETF	Global Equity	LONG
31	QUS	0.0211	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	FLAT
16	CATH	0.0210	Global X S&P 500 Catholic Values ETF	Global Equity	FLAT

26	XLG	0.0210	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	FLAT
18	DGRW	0.0209	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	FLAT
32	SPHQ	0.0208	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	FLAT
7	AOR	0.0207	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	FLAT
47	OMFL	0.0203	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	FLAT
12	MOAT	0.0197	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	FLAT
6	AOM	0.0194	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	FLAT
2	QAI	0.0186	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	FLAT
5	AOK	0.0178	iShares Core 30/70 Conservative Allocat...	Asset Allocation	FLAT

Cluster 6 – U.S. Large Cap Growth

Members: 44 **Var explained:** 0.883 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +54.9%)
- Latest sell_cross: 2026-07-08
- Current 252d return: +27.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-08 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Technology	23	0.5198
U.S. Large Cap Growth	11	0.2593
Growth Equity	2	0.0464
U.S. Equity	2	0.0463
U.S. Mega Cap Growth	1	0.0234
U.S. Equal Weight Equity	1	0.0230
Options Income	1	0.0213
U.S. Small Cap Technology	1	0.0213
Trend-Following Equity	1	0.0209
U.S. Trend-Following Equity	1	0.0183

Replication basket – QQQ (cor=0.98) basket lead at lag 1: -0.025

ticker	rep_wt	name	category	etf_signal
QQQ	1.018	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	IGM	0.0239	iShares Expanded Tech Sector ETF	Technology	FLAT
7	QQQ	0.0238	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	FLAT
23	IYW	0.0238	iShares U.S. Technology ETF	Technology	FLAT
25	ONEQ	0.0238	Fidelity Nasdaq Composite Index ETF	U.S. Equity	FLAT
29	VGT	0.0238	Vanguard Information Tech ETF	Technology	FLAT
31	FTEC	0.0238	Fidelity MSCI Information Technology In...	Technology	FLAT
34	ILCG	0.0237	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
1	XLK	0.0236	State Street Technology Select Sector S...	Technology	FLAT
22	IXN	0.0236	iShares Global Tech ETF	Technology	FLAT
35	IUSG	0.0236	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	FLAT
17	XNTK	0.0235	State Street SPDR NYSE Technology ETF	Technology	FLAT
24	NULG	0.0235	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
27	IWF	0.0235	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	FLAT
28	IVW	0.0235	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	FLAT
30	VOOG	0.0235	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	FLAT
32	VONG	0.0235	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	FLAT
37	SPYG	0.0235	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	FLAT
38	VUG	0.0235	Vanguard Growth ETF	U.S. Large Cap Growth	FLAT
11	QTEC	0.0234	First Trust NASDAQ-100-Technology Secto...	Technology	FLAT

26	IWY	0.0234	iShares Russell Top 200 Growth ETF	Growth Equity	FLAT
33	MGK	0.0234	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	FLAT
16	FXL	0.0233	First Trust Technology AlphaDEX	Technology	FLAT
36	SCHG	0.0233	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
6	AIQ	0.0232	Global X Artificial Intelligence & Tech...	Technology	FLAT
20	PWB	0.0231	Invesco Large Cap Growth ETF	Growth Equity	FLAT
10	QQEW	0.0230	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	FLAT
14	XT	0.0228	iShares Future Exponential Technologies...	Technology	FLAT
13	TDIV	0.0226	First Trust NASDAQ Technology Dividend ...	Technology	FLAT
39	SPHB	0.0225	Invesco S&P 500 High Beta ETF	U.S. Equity	FLAT
3	SMH	0.0224	VanEck Semiconductor ETF	Technology	FLAT
2	SOXX	0.0222	iShares PHLX SOX Semiconductor Sector I...	Technology	FLAT
44	QTUM	0.0222	Defiance Quantum ETF	Technology	LONG
5	ROBO	0.0221	ROBO Global Robotics and Automation Ind...	Technology	FLAT
4	BOTZ	0.0220	Global X Robotics & Artificial Intellig...	Technology	FLAT
15	XSD	0.0218	State Street SPDR S&P Semiconductor ETF	Technology	FLAT
8	NXTG	0.0217	First Trust Indxx NextG ETF	Technology	FLAT
12	PSI	0.0217	Invesco Semiconductors ETF	Technology	FLAT
42	FTXL	0.0217	First Trust Nasdaq Semiconductor ETF	Technology	FLAT
19	QYLD	0.0213	Global X NASDAQ 100 Covered Call ETF	Options Income	FLAT
41	PSCT	0.0213	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	FLAT
43	PTF	0.0213	Invesco Dorsey Wright Technology Moment...	Technology	LONG
18	PTNQ	0.0209	Pacer Trendpilot 100 ETF	Trend-Following Equity	FLAT
9	PRNT	0.0193	3D Printing (The) ETF	Technology	FLAT
40	PTLC	0.0183	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	FLAT

Cluster 7 – U.S. Extended Market Equity

Members: 41 **Var explained:** 0.787 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +25.9%)
- Latest sell_cross: 2026-07-16
- Current 252d return: +7.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-16 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Technology	11	0.2649
U.S. Mid Cap Growth	3	0.0803
U.S. Small Cap Growth	3	0.0790
U.S. Momentum Equity	3	0.0712
Communication Services	3	0.0688
Growth Equity	2	0.0518
IPO Equity	2	0.0509
Convertible Bond	2	0.0502
Momentum Equity	2	0.0489
Financials	2	0.0487
Consumer Discretionary	2	0.0463
U.S. Extended Market Equity	1	0.0267
U.S. Small Cap Momentum Equity	1	0.0240
U.S. Mid Cap Momentum Equity	1	0.0239
Industrials	1	0.0227
International IPO Equity	1	0.0215
Israel Equity	1	0.0202

Replication basket – VXF (cor=0.97) basket lead at lag 1: 0.013

ticker	rep_wt	name	category	etf_signal
VXF	0.987	Vanguard Extended Market ETF	U.S. Extended Market ...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
25	IWP	0.0268	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	FLAT
30	VOT	0.0268	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	FLAT
35	VBK	0.0268	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	FLAT
31	IMCG	0.0267	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	FLAT
37	VXF	0.0267	Vanguard Extended Market ETF	U.S. Extended Market Equ...	FLAT
36	ISCG	0.0263	iShares Morningstar Small-Cap Growth ETF	Growth Equity	FLAT
33	IWO	0.0262	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	FLAT
34	VTWG	0.0261	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	FLAT
21	FPX	0.0259	First Trust US Equity Opportunities ETF	IPO Equity	FLAT
19	FTC	0.0257	First Trust Large Cap Growth AlphaDEX F...	Technology	FLAT
23	RPG	0.0255	Invesco S&P 500 Pure Growth ETF	Growth Equity	FLAT
22	CWB	0.0254	State Street SPDR Bloomberg Convertible...	Convertible Bond	FLAT
20	PDP	0.0252	Invesco Dorsey Wright Momentum ETF	Momentum Equity	FLAT
13	IPO	0.0250	Renaissance IPO ETF	IPO Equity	FLAT
18	FINX	0.0250	Global X FinTech ETF	Financials	FLAT
39	VFMO	0.0250	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	FLAT
8	ARKQ	0.0249	ARK Autonomous Technology & Robotics ETF	Technology	FLAT
15	FDN	0.0249	First Trust DJ Internet Index Fund	Technology	FLAT
26	ICVT	0.0249	iShares Convertible Bond ETF	Convertible Bond	FLAT
6	ARKK	0.0245	ARK Innovation ETF	Technology	FLAT
7	ARKW	0.0245	ARK Next Generation Internet ETF	Technology	FLAT
3	SKYY	0.0244	First Trust Cloud Computing ETF	Technology	FLAT
14	PNQI	0.0244	Invesco Nasdaq Internet ETF	Technology	FLAT
11	IBUY	0.0243	Amplify Online Retail ETF	Consumer Discretionary	FLAT
32	DWAS	0.0240	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	FLAT
24	XMMO	0.0239	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	FLAT
16	IPAY	0.0238	Amplify Digital Payments ETF	Financials	FLAT
10	FV	0.0237	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	FLAT
2	IGV	0.0234	iShares Expanded Tech-Software Sector E...	Technology	FLAT
27	MTUM	0.0234	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	FLAT
28	VOX	0.0233	Vanguard Communication Services ETF	Communication Services	FLAT
29	FCOM	0.0233	Fidelity MSCI Communication Services In...	Communication Services	FLAT
12	CIBR	0.0230	First Trust NASDAQ Cybersecurity ETF	Technology	FLAT
38	QMOM	0.0228	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	FLAT
5	HACK	0.0227	Amplify Cybersecurity ETF	Technology	FLAT
41	PRN	0.0227	Invesco Dorsey Wright Industrials Momen...	Industrials	FLAT

4	BLOK	0.0224	Amplify Blockchain Technology ETF	Technology	FLAT
1	XLC	0.0222	State Street Communication Services Sel...	Communication Services	FLAT
9	ESPO	0.0220	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
17	FPXI	0.0215	First Trust International Equity Opport...	International IPO Equity	FLAT
40	EIS	0.0202	iShares MSCI Israel ETF	Israel Equity	FLAT

Cluster 8 – Emerging Markets Equity

Members: 25 **Var explained:** 0.871 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +57.0%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +23.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Emerging Markets Equity	19	0.7728

Asia Equity	2	0.0815
Asia ex-Japan Equity	1	0.0418
Taiwan Equity	1	0.0362
South Korea Equity	1	0.0340
Global Dividend Equity	1	0.0337

Replication basket – IEMG (cor=0.99) basket lead at lag 1: -0.034

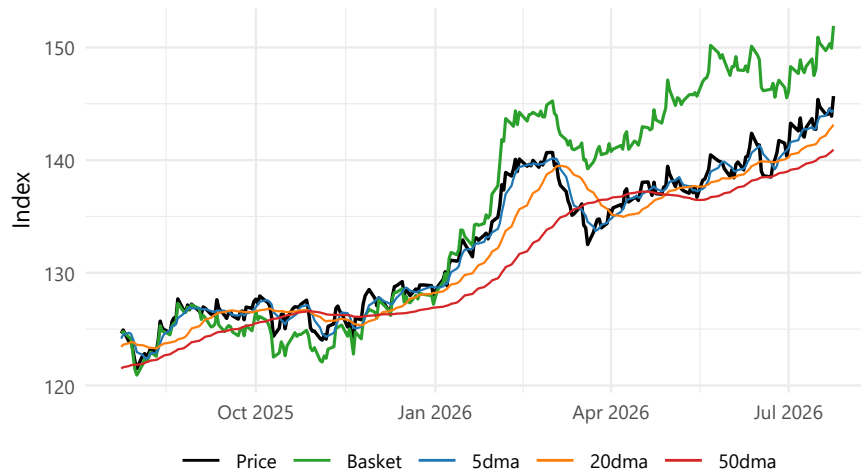
ticker	rep_wt	name	category	etf_signal
IEMG	0.919	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	IEMG	0.0425	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	FLAT
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	FLAT
20	ESGE	0.0423	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	FLAT
23	SPEM	0.0423	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	FLAT
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	FLAT
16	GEM	0.0422	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	FLAT
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	FLAT
18	XSOE	0.0420	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	FLAT
15	EMGF	0.0419	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	FLAT
5	AAXJ	0.0418	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	FLAT
11	EEMA	0.0414	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	FLAT
14	GMF	0.0412	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	FLAT
12	PXH	0.0409	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	FLAT
17	FNDE	0.0409	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	LONG
19	EEMV	0.0405	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	FLAT
10	AIA	0.0403	iShares Asia 50 ETF	Asia Equity	LONG
8	DGS	0.0402	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	FLAT
7	DEM	0.0397	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	FLAT
6	EWX	0.0392	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	FLAT
4	FEM	0.0389	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	FLAT
13	DVYE	0.0364	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	LONG
9	EWI	0.0362	iShares MSCI Taiwan ETF	Taiwan Equity	FLAT
25	FEMS	0.0349	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	FLAT
1	EWY	0.0340	iShares MSCI South Korea ETF	South Korea Equity	FLAT
24	SDIV	0.0337	Global X SuperDividend ETF	Global Dividend Equity	LONG

Cluster 9 – U.S. Dividend Equity**Members:** 25 **Var explained:** 0.869 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-04-20 – filter **PASS** (252d at date = +20.8%)
- Latest sell_cross: 2026-06-24
- Current 252d return: +16.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-24 more recent than last effective buy 2026-04-20)

**Category breakdown**

category	n_members	abs_weight
U.S. Dividend Equity	13	0.5282
U.S. Low Volatility Equity	4	0.1562
Value Equity	3	0.1201
Dividend Equity	2	0.0792
U.S. Defensive Equity	1	0.0404
U.S. Quality / Factor Equity	1	0.0390
Global Equity	1	0.0368

Replication basket – SCHD (cor=0.96) basket lead at lag 1: 0.004

ticker	rep_wt	name	category	etf_signal
SCHD	0.893	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	SDY	0.0418	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
15	DVY	0.0417	iShares Select Dividend ETF	U.S. Dividend Equity	FLAT
2	FVD	0.0415	First Trust VL Dividend	U.S. Dividend Equity	LONG
10	SCHD	0.0413	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
14	SDOG	0.0413	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	FLAT
20	SPYD	0.0412	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	FLAT
5	DHS	0.0411	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	FLAT
11	FTA	0.0411	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	LONG
19	ONEY	0.0410	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	FLAT
18	SPHD	0.0409	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
6	CDC	0.0404	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	FLAT
4	FDL	0.0402	First Trust Morningstar ETF	U.S. Dividend Equity	LONG
25	LGLV	0.0400	State Street SPDR US Large Cap Low Vola...	U.S. Low Volatility Equi...	LONG
8	LVHD	0.0399	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	LONG
3	PWV	0.0398	Invesco Large Cap Value ETF	Value Equity	LONG
9	HDV	0.0394	iShares Core High Dividend ETF	U.S. Dividend Equity	LONG
16	RDIV	0.0394	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	FLAT
12	PEY	0.0393	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	FLAT
13	DIV	0.0391	Global X Super Dividend ETF	U.S. Dividend Equity	LONG
17	RPV	0.0391	Invesco S&P 500 Pure Value ETF	Value Equity	FLAT
21	FTCS	0.0390	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	LONG
22	OUSA	0.0384	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
23	SPLV	0.0381	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	LONG
24	USMV	0.0381	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	FLAT
1	ACWV	0.0368	iShares MSCI Global Min Vol Factor ETF	Global Equity	LONG

Cluster 10 – High Yield Bond

Members: 21 **Var explained:** 0.765 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +7.3%)

- Latest sell_cross: 2026-07-07
- Current 252d return: +3.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-17)



Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.5157
Preferred Stock	7	0.3058
Emerging Markets Fixed Income	4	0.1785

Replication basket – HYG (cor=0.95) basket lead at lag 1: 0.047

ticker	rep_wt	name	category	etf_signal
HYG	0.917	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
13 JNK	0.0527	State Street SPDR Bloomberg High Yield ...	High Yield Bond	FLAT

1	HYG	0.0526	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	FLAT
19	HYLE	0.0524	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	FLAT
18	BBHY	0.0523	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	FLAT
16	SHYG	0.0521	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	FLAT
14	SJNK	0.0520	State Street SPDR Bloomberg Short Term ...	High Yield Bond	FLAT
17	FALN	0.0517	iShares Fallen Angels USD Bond ETF	High Yield Bond	FLAT
15	ANGL	0.0516	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	FLAT
6	HYS	0.0511	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	FLAT
9	PFF	0.0479	iShares Preferred and Income Securities...	Preferred Stock	FLAT
3	HYLS	0.0472	First Trust Tactical High Yield ETF	High Yield Bond	FLAT
2	EMB	0.0468	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
21	VWOB	0.0467	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	FLAT
11	PFXF	0.0460	VanEck Preferred Securities ex Financia...	Preferred Stock	FLAT
20	PCY	0.0458	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	FLAT
7	PGX	0.0446	Invesco Preferred ETF	Preferred Stock	FLAT
10	PSK	0.0441	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT
4	FPE	0.0433	First Trust Preferred Securities and In...	Preferred Stock	FLAT
5	PGF	0.0431	Invesco Financial Preferred ETF	Preferred Stock	FLAT
12	HYEM	0.0392	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	FLAT
8	VRP	0.0368	Invesco Variable Rate Preferred ETF	Preferred Stock	FLAT

Cluster 11 – Healthcare

Members: 17 **Var explained:** 0.758 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +13.8%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +24.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9418
U.S. Small Cap Healthcare	1	0.0582

Replication basket – VHT (cor=0.93) basket lead at lag 1: -0.004

ticker	rep_wt	name	category	etf_signal
VHT	1.092	Vanguard Health Care ETF	Healthcare	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	VHT	0.0655	Vanguard Health Care ETF	Healthcare	LONG
16	FHLC	0.0654	Fidelity MSCI Health Care Index ETF	Healthcare	LONG
9	FXH	0.0648	First Trust Health Care AlphaDEX	Healthcare	FLAT
14	RSPH	0.0641	Invesco S&P 500 Equal Weight Health Car...	Healthcare	FLAT
13	IYH	0.0639	iShares U.S. Healthcare ETF	Healthcare	LONG
10	IXJ	0.0624	iShares Global Healthcare ETF	Healthcare	FLAT

1	XLV	0.0622	State Street Health Care Select Sector ...	Healthcare	LONG
3	IBB	0.0604	iShares Biotechnology ETF	Healthcare	FLAT
6	FBT	0.0597	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0582	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	LONG
7	XHE	0.0573	State Street SPDR S&P Health Care Equip...	Healthcare	FLAT
12	IHI	0.0567	iShares U.S. Medical Devices ETF	Healthcare	FLAT
17	PPH	0.0545	VanEck Pharmaceutical ETF	Healthcare	LONG
5	PTH	0.0538	Invesco Dorsey Wright Healthcare Moment...	Healthcare	LONG
4	XBI	0.0532	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
2	ARKG	0.0506	ARK Genomic Revolution ETF	Healthcare	FLAT
11	IHF	0.0472	iShares U.S. Health Care Providers ETF	Healthcare	LONG

Cluster 12 – Materials

Members: 17 **Var explained:** 0.687 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +48.9%)
- Latest sell_cross: 2026-06-09
- Current 252d return: +26.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-09 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Materials	9	0.5590
MLP / Energy Infrastructure	4	0.2178
Energy	1	0.0644
Consumer Staples	1	0.0597
Norway Equity	1	0.0556
Industrial Metals	1	0.0434

Replication basket – AMLP+URA+GUNR (cor=0.98) basket lead at lag 1: 0.046

ticker	rep_wt	name	category	etf_signal
AMLP	0.139	Alerian MLP ETF	MLP / Energy Infrastr...	LONG
URA	0.094	Global X Uranium ETF	Materials	FLAT
GUNR	0.776	FlexShares Global Upstream Natural Re...	Materials	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	GNR	0.0691	State Street SPDR S&P Global Natural Re...	Materials	FLAT
10	GUNR	0.0688	FlexShares Global Upstream Natural Reso...	Materials	FLAT
17	NANR	0.0668	State Street SPDR S&P North American Na...	Materials	LONG
16	IGE	0.0644	iShares North American Natural Resource...	Energy	LONG
9	MXI	0.0634	iShares Global Materials ETF	Materials	FLAT
12	PICK	0.0631	iShares MSCI Global Select Metals & Min...	Materials	FLAT
3	XME	0.0614	State Street SPDR S&P Metals & Mining E...	Materials	FLAT
13	SLX	0.0607	VanEck Steel ETF	Materials	FLAT
5	MOO	0.0597	VanEck Agribusiness ETF	Consumer Staples	LONG
2	COPX	0.0589	Global X Copper Miners ETF	Materials	FLAT
8	MLPX	0.0565	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
14	NORW	0.0556	Global X MSCI Norway ETF	Norway Equity	FLAT
4	EMLP	0.0550	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	LONG
6	AMLP	0.0538	Alerian MLP ETF	MLP / Energy Infrastruct...	LONG
7	MLPA	0.0525	Global X MLP ETF	MLP / Energy Infrastruct...	LONG
1	URA	0.0467	Global X Uranium ETF	Materials	FLAT
15	DBB	0.0434	Invesco DB Base Metals Fund	Industrial Metals	FLAT

Cluster 13 – China Equity**Members:** 15 **Var explained:** 0.797 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-04-15 – filter **PASS** (252d at date = +24.3%)
- Latest sell_cross: 2026-05-20
- Current 252d return: -2.5% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-04-15)

**Category breakdown**

category	n_members	abs_weight
China Equity	13	0.8765
Emerging Markets Equity	1	0.0644
Hong Kong Equity	1	0.0591

Replication basket – KWEB (cor=0.92) basket lead at lag 1: 0.011

ticker	rep_wt	name	category	etf_signal
KWEB	0.562	KraneShares CSI China Internet ETF	China Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	CXSE	0.0739	WisdomTree China ex-State-Owned Enterpr...	China Equity	FLAT
9	GXC	0.0737	State Street SPDR S&P China ETF	China Equity	FLAT
1	MCHI	0.0731	iShares MSCI China ETF	China Equity	FLAT
5	CQQQ	0.0712	Invesco China Technology ETF	China Equity	FLAT
2	FXI	0.0711	iShares China Large-Cap ETF	China Equity	FLAT
7	CHIQ	0.0705	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0680	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0668	Xtrackers Harvest CSI 300 China A-Share...	China Equity	FLAT
8	CNYA	0.0660	iShares MSCI China A ETF	China Equity	FLAT
13	KGRN	0.0653	KraneShares MSCI China Clean Technology...	China Equity	FLAT
3	EMQQ	0.0644	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
10	KBA	0.0596	KraneShares Boseri MSCI China A 50 Conn...	China Equity	FLAT
12	EWI	0.0591	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	LONG
14	CNXT	0.0590	VanEck ChiNext Innovators ETF	China Equity	FLAT
15	KURE	0.0582	KraneShares MSCI All China Health Care ...	China Equity	LONG

Cluster 14 – Financials

Members: 14 **Var explained:** 0.873 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +19.9%)
- Latest sell_cross: 2026-05-11
- Current 252d return: +9.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-11 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Financials	13	0.9382
Global Equity	1	0.0618

Replication basket – XLF (cor=0.96) basket lead at lag 1: 0.012

ticker	rep_wt	name	category	etf_signal
XLF	1.095	State Street Financial Select Sector ...	Financials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
10	VFH	0.0756	Vanguard Financials ETF	Financials	FLAT
11	FNCL	0.0756	Fidelity MSCI Financials Index ETF	Financials	LONG
7	IYF	0.0753	iShares U.S. Financial ETF	Financials	FLAT
5	FXO	0.0752	First Trust Financials AlphaDEX	Financials	LONG
8	IYG	0.0749	iShares U.S. Financial Services ETF	Financials	FLAT
1	XLF	0.0742	State Street Financial Select Sector SP...	Financials	LONG

9	KBWB	0.0734	Invesco KBW Bank ETF	Financials	FLAT
3	KBE	0.0712	State Street SPDR S&P Bank ETF	Financials	FLAT
14	KCE	0.0708	State Street SPDR S&P Capital Markets E...	Financials	LONG
6	IAT	0.0706	iShares U.S. Regional Banks ETF	Financials	LONG
13	IAI	0.0696	iShares U.S. Broker-Dealers & Securitie...	Financials	LONG
2	KRE	0.0687	State Street SPDR S&P Regional Banking ...	Financials	FLAT
4	KIE	0.0631	State Street SPDR S&P Insurance ETF	Financials	LONG
12	PSP	0.0618	Invesco Global Listed Private Equity ETF	Global Equity	FLAT

Cluster 15 – Real Estate

Members: 14 **Var explained:** 0.961 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +15.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ (cor=0.99) basket lead at lag 1: -0.002

ticker	rep_wt	name	category	etf_signal
VNQ	0.97	Vanguard Real Estate ETF	Real Estate	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

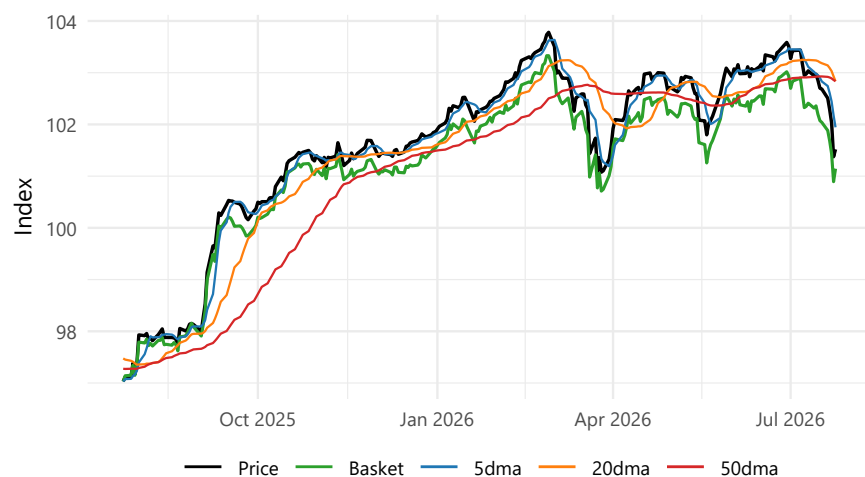
	ticker	PC_wt	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	FLAT
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	LONG
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	FLAT
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	LONG
10	USRT	0.0724	iShares Core U.S. REIT ETF	Real Estate	FLAT
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	FLAT
13	PSR	0.0722	Invesco Active U.S. Real Estate Fund	Real Estate	FLAT
1	XLRE	0.0719	State Street Real Estate Select Sector ...	Real Estate	LONG
5	ICF	0.0719	iShares Select U.S. REIT ETF	Real Estate	FLAT
7	REET	0.0716	iShares Global REIT ETF	Real Estate	FLAT
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	LONG
4	REZ	0.0697	iShares Residential and Multisector Rea...	Real Estate	FLAT
14	NURE	0.0688	Nuveen Short-Term REIT ETF	Real Estate	LONG
12	INDS	0.0683	Pacer Industrial Real Estate ETF	Real Estate	LONG

Cluster 16 – Municipal Bond

Members: 13 **Var explained:** 0.820 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-28 – filter **PASS** (252d at date = +5.9%)
- Latest sell_cross: 2026-07-10
- Current 252d return: +4.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-10 more recent than last effective buy 2026-05-28)



Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB (cor=0.95) basket lead at lag 1: 0.179

ticker	rep_wt	name	category	etf_signal
MUB	0.904	iShares National Muni Bond ETF	Municipal Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	FLAT
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	FLAT
6	CMF	0.0801	iShares California Muni Bond ETF	Municipal Bond	FLAT
9	TFI	0.0801	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	FLAT
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	FLAT
1	FMB	0.0794	First Trust Managed Municipal ETF	Municipal Bond	FLAT
7	NYF	0.0793	iShares New York Muni Bond ETF	Municipal Bond	FLAT

5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	FLAT
4	MUNI	0.0770	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	FLAT
2	HYD	0.0747	VanEck High Yield Muni ETF	Municipal Bond	FLAT
3	HYMB	0.0721	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	FLAT
12	SUB	0.0695	iShares Short-Term National Muni Bond E...	Municipal Bond	FLAT
10	SHM	0.0668	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	FLAT

Cluster 17 – Gold

Members: 12 **Var explained:** 0.863 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-12 – filter **PASS** (252d at date = +75.1%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +21.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-05-12)



Category breakdown

category	n_members	abs_weight
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Materials	5	0.4157
Gold	3	0.2502
Precious Metals	2	0.1724
Silver	2	0.1618

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.011

ticker	rep_wt	name	category	etf_signal
GLD	0.017	SPDR Gold Shares	Gold	FLAT
IAU	0.754	iShares Gold Trust Shares	Gold	FLAT
SLV	0.412	iShares Silver Trust	Silver	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	DBP	0.0865	Invesco DB Precious Metals Fund	Precious Metals	FLAT
7	GLTR	0.0858	abrdn Physical Precious Metals Basket S...	Precious Metals	FLAT
1	GDX	0.0840	VanEck Gold Miners ETF	Materials	FLAT
2	GDXJ	0.0840	VanEck Junior Gold Miners ETF	Materials	FLAT
3	GLD	0.0834	SPDR Gold Shares	Gold	FLAT
9	IAU	0.0834	iShares Gold Trust Shares	Gold	FLAT
10	SGOL	0.0833	abrdn Physical Gold Shares ETF	Gold	FLAT
12	GOAU	0.0831	US Global GO Gold and Precious Metal Mi...	Materials	FLAT
6	SIL	0.0828	Global X Silver Miners ETF	Materials	FLAT
5	SILJ	0.0818	Amplify Junior Silver Miners ETF	Materials	FLAT
4	SLV	0.0810	iShares Silver Trust	Silver	LONG
8	SIVR	0.0808	abrdn Physical Silver Shares ETF	Silver	LONG

Cluster 18 – Consumer Staples

Members: 11 **Var explained:** 0.744 **Signal:** LONG since 2026-06-26 (19 days, -0.2%)

Signal logic:

- Latest buy_cross: 2026-06-26 – filter **PASS** (252d at date = +9.9%)
- Latest sell_cross: 2026-05-29
- Current 252d return: +5.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-26 > last sell at 2026-05-29)



Category breakdown

category	n_members	abs_weight
Consumer Staples	6	0.5736
Utilities	4	0.3514
U.S. Small Cap Consumer Staples	1	0.0749

Replication basket – XLU+XLP (cor=0.99) basket lead at lag 1: -0.018

ticker	rep_wt	name	category	etf_signal
XLU	0.363	State Street Utilities Select Sector ...	Utilities	FLAT
XLP	0.588	State Street Consumer Staples Select ...	Consumer Staples	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	VDC	0.0983	Vanguard Consumer Staples ETF	Consumer Staples	LONG
6	FSTA	0.0983	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0975	State Street Consumer Staples Select Se...	Consumer Staples	LONG
4	IYK	0.0950	iShares U.S. Consumer Staples ETF	Consumer Staples	LONG

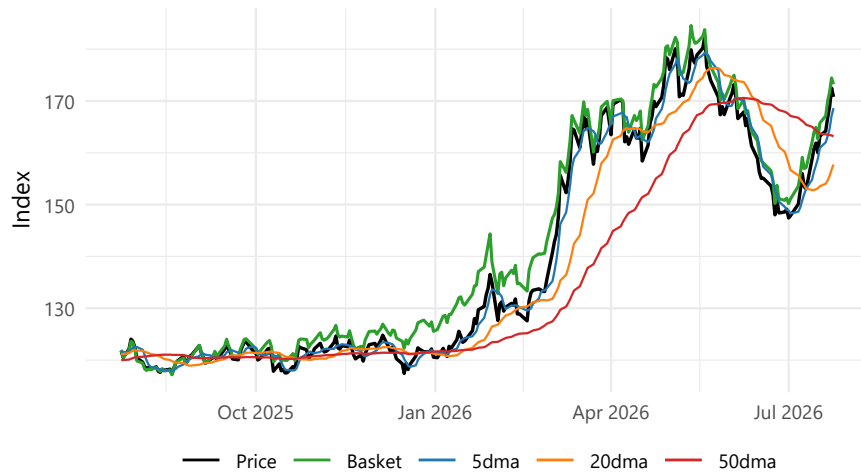
3	KXI	0.0947	iShares Global Consumer Staples ETF	Consumer Staples	LONG
11	PBJ	0.0898	Invesco Food & Beverage ETF	Consumer Staples	LONG
7	IDU	0.0889	iShares U.S. Utilities ETF	Utilities	LONG
9	FUTY	0.0877	Fidelity MSCI Utilities Index ETF	Utilities	FLAT
8	VPU	0.0876	Vanguard Utilities ETF	Utilities	FLAT
2	XLU	0.0873	State Street Utilities Select Sector SP...	Utilities	FLAT
10	PSCC	0.0749	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	LONG

Cluster 19 – Broad Commodities

Members: 11 **Var explained:** 0.771 **Signal:** LONG since 2026-07-22 (2 days, 1.2%)

Signal logic:

- Latest buy_cross: 2026-07-22 – filter **PASS** (252d at date = +38.4%)
- Latest sell_cross: 2026-05-27
- Current 252d return: +40.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-07-22 > last sell at 2026-05-27)



Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6174
Oil	2	0.1980
Energy Commodities	1	0.1019
Agricultural Commodities	1	0.0498
Natural Gas	1	0.0328

Replication basket – PDBC (cor=0.98) basket lead at lag 1: 0.077

ticker	rep_wt	name	category	etf_signal
PDBC	1.122	Invesco Optimum Yield Diversified Com...	Broad Commodities	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	DBC	0.1056	Invesco DB Commodity Index Tracking Fund	Broad Commodities	LONG
6	GSG	0.1056	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	LONG
8	PDBC	0.1056	Invesco Optimum Yield Diversified Commo...	Broad Commodities	LONG
9	COMT	0.1051	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	LONG
10	DBE	0.1019	Invesco DB Energy Fund	Energy Commodities	LONG
1	USO	0.0995	United States Oil Fund	Oil	LONG
5	DBO	0.0984	Invesco DB Oil Fund	Oil	LONG
7	DJP	0.0983	iPath Bloomberg Commodity Index Total R...	Broad Commodities	LONG
4	FTGC	0.0972	First Trust Global Tactical Commodity S...	Broad Commodities	LONG
3	DBA	0.0498	Invesco DB Agriculture Fund	Agricultural Commodities	LONG
11	UNG	0.0328	United States Natural Gas Fund LP	Natural Gas	FLAT

Cluster 20 – India Equity

Members: 10 **Var explained:** 0.405 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +4.7%)
- Latest sell_cross: 2026-07-13
- Current 252d return: -15.8% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-13 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
India Equity	3	0.4681
Malaysia Equity	1	0.1124
Indonesia Equity	1	0.1039
Philippines Equity	1	0.1013
Vietnam Equity	1	0.0747
Turkey Equity	1	0.0611
U.S. Dividend Equity	1	0.0548
Platinum	1	0.0237

Replication basket – NOBL+INDA+PPLT (cor=0.91) basket lead at lag 1: -0.006

ticker	rep_wt	name	category	etf_signal
NOBL	0.080	ProShares S&P 500 Dividend Aristocrat...	U.S. Dividend Equity	FLAT
INDA	0.660	iShares MSCI India ETF	India Equity	FLAT
PPLT	0.029	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	INDA	0.1574	iShares MSCI India ETF	India Equity	FLAT
2	INDY	0.1557	iShares India 50 ETF	India Equity	FLAT
3	EPI	0.1550	WisdomTree India Earnings Fund	India Equity	FLAT
8	EWM	0.1124	iShares MSCI Malaysia Index Fund	Malaysia Equity	FLAT
6	EIDO	0.1039	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
7	EPHE	0.1013	iShares MSCI Philippines ETF	Philippines Equity	FLAT
10	VNM	0.0747	VanEck Vietnam ETF	Vietnam Equity	FLAT
9	TUR	0.0611	iShares MSCI Turkey ETF	Turkey Equity	LONG
4	NOBL	0.0548	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
5	PPLT	0.0237	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Cluster 21 – Inflation-Protected Fixed Income

Members: 10 **Var explained:** 0.905 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-27 – filter **PASS** (252d at date = +4.8%)
- Latest sell_cross: 2026-07-02
- Current 252d return: +0.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-02 more recent than last effective buy 2026-05-27)



Category breakdown

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – VTIP+SCHP (cor=0.99) basket lead at lag 1: -0.013

ticker	rep_wt	name	category	etf_signal
VTIP	0.269	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT
SCHP	0.760	Schwab U.S. TIPS ETF	Inflation-Protected F...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	TDTF	0.1035	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	FLAT
6	TIPX	0.1034	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	FLAT
3	TIP	0.1030	iShares TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
8	SCHP	0.1023	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	FLAT
5	TDTT	0.1008	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.1000	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	FLAT

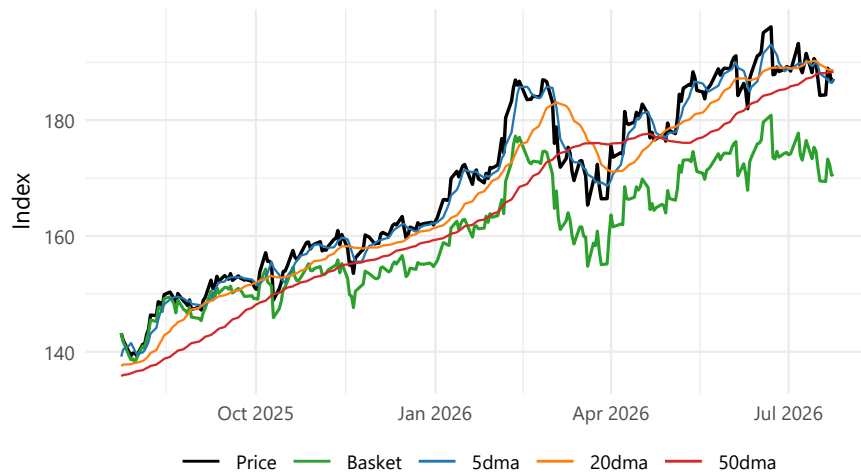
9	STIP	0.0998	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
2	STPZ	0.0996	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0990	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
1	LTPZ	0.0885	Pimco 15 Year U.S. TIPS Index Exchange-...	Inflation-Protected Fixe...	FLAT

Cluster 22 – Japan Equity

Members: 9 **Var explained:** 0.888 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +57.2%)
- Latest sell_cross: 2026-07-17
- Current 252d return: +30.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-17 more recent than last effective buy 2026-04-09)



Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7804

Asia-Pacific Equity	2	0.2196
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Replication basket – EWJ (cor=0.97) basket lead at lag 1: -0.043

ticker	rep_wt	name	category	etf_signal
EWJ	0.91	iShares MSCI Japan Index Fund	Japan Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	EWJ	0.1142	iShares MSCI Japan Index Fund	Japan Equity	FLAT
8	BBJP	0.1142	JPMorgan BetaBuilders Japan ETF	Japan Equity	FLAT
6	DBJP	0.1118	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	LONG
9	HEWJ	0.1118	iShares Currency Hedged MSCI Japan ETF	Japan Equity	LONG
3	IPAC	0.1114	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	FLAT
5	FLJH	0.1110	Franklin FTSE Japan Hedged ETF	Japan Equity	FLAT
2	DXJ	0.1096	WisdomTree Japan Hedged Equity Fund	Japan Equity	FLAT
4	VPL	0.1082	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	FLAT
7	EWJV	0.1078	iShares MSCI Japan Value ETF	Japan Equity	LONG

Cluster 23 – Bank Loan / Floating Rate

Members: 9 **Var explained:** 0.556 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +6.5%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +3.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3621
Interest Rate Hedged Fixed Income	3	0.3244
Floating Rate Fixed Income	3	0.3135

Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.047

ticker	rep_wt	name	category	etf_signal
FLOT	0.291	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	LONG
BKLN	0.214	Invesco Senior Loan ETF	Bank Loan / Floating ...	FLAT
SRLN	0.302	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	LONG
FLRN	0.128	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	LONG
FLTR	0.034	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
2 SRLN	0.1250	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	LONG

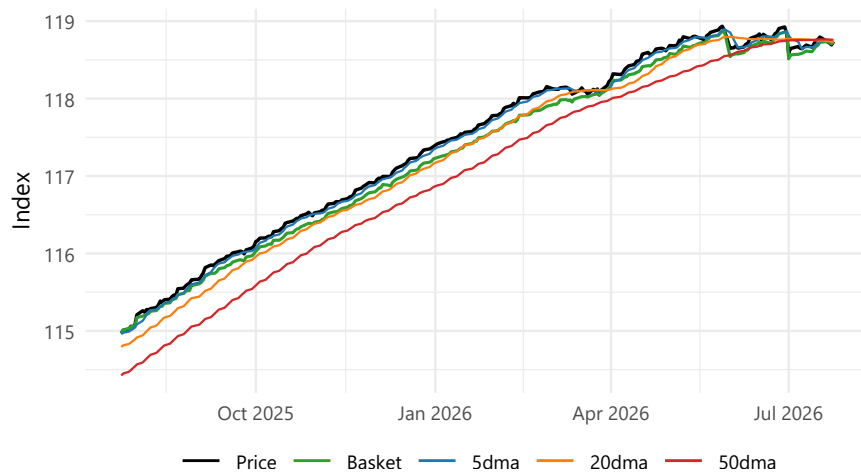
3	BKLN	0.1208	Invesco Senior Loan ETF	Bank Loan / Floating Rate	FLAT
1	FTSL	0.1163	First Trust Senior Loan Fund	Bank Loan / Floating Rate	FLAT
5	LQDH	0.1144	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	FLAT
7	FLOT	0.1084	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	LONG
6	IGBH	0.1056	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	FLAT
4	IGHG	0.1045	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	FLAT
8	FLRN	0.1039	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	LONG
9	FLTR	0.1011	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	FLAT

Cluster 24 – Cash / Ultra-Short Fixed Income

Members: 9 Var explained: 0.456 Signal: FLAT

Signal logic:

- Latest buy_cross: 2022-10-28 – filter **BLOCKED** (252d at date = -0.5%; needs > 0)
- Latest effective buy: 2022-11-29 – delayed re-arm from blocked cross (252d at fire = +0.0%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +3.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2022-11-29)



Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – SHV+MINT+GBIL+ICSH (cor=0.85) basket lead at lag 1: 0.065

ticker	rep_wt	name	category	etf_signal
SHV	0.275	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	LONG
MINT	0.172	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	LONG
GBIL	0.122	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	LONG
ICSH	0.403	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	SHV	0.1309	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	LONG
6	ICSH	0.1297	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	LONG
5	GSY	0.1162	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	FLAT
2	MINT	0.1127	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	LONG
3	FTSM	0.1126	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	LONG
4	NEAR	0.1107	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	FLAT
8	BIL	0.1019	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	LONG
1	VNLA	0.0953	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	FLAT
9	GBIL	0.0901	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	LONG

Cluster 25 – International Fixed Income

Members: 6 **Var explained:** 0.742 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-29 – filter **PASS** (252d at date = +0.6%)
- Latest sell_cross: 2026-07-23
- Current 252d return: -4.7% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-05-29)



Category breakdown

category	n_members	abs_weight
Currency	4	0.6409
International Fixed Income	2	0.3591

Replication basket – BWX (cor=0.93) basket lead at lag 1: 0.025

ticker	rep_wt	name	category	etf_signal
BWX	0.751	State Street SPDR Bloomberg Internati...	International Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	FLAT
1	BWX	0.1790	State Street SPDR Bloomberg Internation...	International Fixed Inco...	FLAT
4	FXE	0.1702	Invesco CurrencyShares Euro Currency Tr...	Currency	FLAT
5	FXF	0.1650	Invesco CurrencyShares Swiss Franc Trust	Currency	FLAT
3	FXB	0.1609	Invesco CurrencyShares British Pound St...	Currency	LONG
6	FXY	0.1448	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 26 – Leveraged / Inverse**Members:** 2 **Var explained:** 0.625 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-07-20 – filter **BLOCKED** (252d at date = -31.3%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell_cross: 2026-07-06
- Current 252d return: -28.6% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-06 more recent than last effective buy 2025-04-08)

**Category breakdown**

category	n_members	abs_weight
Leveraged / Inverse	1	0.5
Currency	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.023

ticker	rep_wt	name	category	etf_signal
SQQQ	0.506	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT

UUP	0.495	Invesco DB USD Index Bullish Fund ETF	Currency	FLAT
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Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	FLAT

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	cluster	theme	entry	days	return	p_sell	nss
FTGC	cluster_19	Broad Commodities	2026-07-17	5	2.1%	0.00	1.00
GSG	cluster_19	Broad Commodities	2026-07-21	3	1.9%	0.00	1.00
DBE	cluster_19	Broad Commodities	2026-07-20	4	5.4%	0.00	1.00
DJP	cluster_19	Broad Commodities	2026-07-22	2	0.1%	0.00	1.00
DBC	cluster_19	Broad Commodities	2026-07-21	3	1.8%	0.00	1.00
DBA	cluster_19	Broad Commodities	2026-07-09	11	1.9%	0.00	1.00
PDBC	cluster_19	Broad Commodities	2026-07-21	3	1.7%	0.00	1.00
DBO	cluster_19	Broad Commodities	2026-07-21	3	6.2%	0.00	1.00
USO	cluster_19	Broad Commodities	2026-07-22	2	3.8%	0.00	1.00
COMT	cluster_19	Broad Commodities	2026-07-23	1	-0.1%	0.00	1.00
SCHD	cluster_9	U.S. Dividend Equity	2026-07-07	13	2.3%	0.00	1.00
PWV	cluster_9	U.S. Dividend Equity	2026-04-02	77	14.6%	0.00	1.00
DIV	cluster_9	U.S. Dividend Equity	2026-06-29	18	3.3%	0.00	1.00
FDL	cluster_9	U.S. Dividend Equity	2026-07-16	6	0.4%	0.00	1.00
HDV	cluster_9	U.S. Dividend Equity	2026-06-25	20	4.4%	0.00	1.00
FTA	cluster_9	U.S. Dividend Equity	2026-05-21	43	6.6%	0.00	1.00
FVD	cluster_9	U.S. Dividend Equity	2026-06-10	30	4.5%	0.06	0.94
SPHD	cluster_9	U.S. Dividend Equity	2026-06-08	32	5.7%	0.09	0.91
LGLV	cluster_9	U.S. Dividend Equity	2026-06-26	19	2.9%	0.10	0.90
LVHD	cluster_9	U.S. Dividend Equity	2026-06-10	30	4.2%	0.27	0.73
ACWV	cluster_9	U.S. Dividend Equity	2026-07-06	14	0.0%	0.19	0.73
FTCS	cluster_9	U.S. Dividend Equity	2026-06-26	19	3.6%	0.33	0.67
OUSA	cluster_9	U.S. Dividend Equity	2026-07-01	16	1.4%	0.32	0.66
SDY	cluster_9	U.S. Dividend Equity	2026-05-20	44	5.1%	0.34	0.66
SPLV	cluster_9	U.S. Dividend Equity	2026-06-11	29	4.5%	0.37	0.63
RWO	cluster_15	Real Estate	2026-06-25	20	4.8%	0.00	1.00
INDS	cluster_15	Real Estate	2026-06-08	32	7.2%	0.00	1.00
FREL	cluster_15	Real Estate	2026-06-25	20	3.6%	0.01	0.99
IYR	cluster_15	Real Estate	2026-06-25	20	3.8%	0.01	0.99
XLRE	cluster_15	Real Estate	2026-06-25	20	3.1%	0.01	0.99
NURE	cluster_15	Real Estate	2026-04-13	71	14.3%	0.53	0.47
KCE	cluster_14	Financials	2026-07-07	13	0.5%	0.14	0.84
XLF	cluster_14	Financials	2026-04-13	71	9.0%	0.18	0.82
FNCL	cluster_14	Financials	2026-04-13	71	9.0%	0.18	0.82
FXO	cluster_14	Financials	2026-06-05	33	8.9%	0.23	0.77
IAI	cluster_14	Financials	2026-07-06	14	-1.3%	0.30	0.63
IAT	cluster_14	Financials	2026-05-20	44	11.9%	0.49	0.51

KIE	cluster_14	Financials	2026-06-10	30	10.4%	0.53	0.47
PBJ	cluster_18	Consumer Staples	2026-07-13	9	-1.4%	0.08	0.92
IYK	cluster_18	Consumer Staples	2026-06-09	31	2.7%	0.29	0.71
PSCC	cluster_18	Consumer Staples	2026-06-10	30	6.1%	0.31	0.69
IDU	cluster_18	Consumer Staples	2026-06-25	20	1.1%	0.52	0.48
XLP	cluster_18	Consumer Staples	2026-07-16	6	-2.0%	0.54	-0.04
KXI	cluster_18	Consumer Staples	2026-07-16	6	-2.2%	0.62	-0.04
FSTA	cluster_18	Consumer Staples	2026-07-16	6	-2.2%	0.71	-0.32
VDC	cluster_18	Consumer Staples	2026-07-16	6	-2.2%	0.75	-0.44
EPP	cluster_1	Developed ex-US Equity	2026-07-15	7	-0.1%	0.00	1.00
BBAX	cluster_1	Developed ex-US Equity	2026-07-14	8	0.9%	0.00	1.00
ILF	cluster_1	Developed ex-US Equity	2026-07-22	2	-2.2%	0.00	1.00
EPOL	cluster_1	Developed ex-US Equity	2026-07-08	12	2.9%	0.00	1.00
HDEF	cluster_1	Developed ex-US Equity	2026-07-09	11	2.0%	0.00	1.00
EFAV	cluster_1	Developed ex-US Equity	2026-07-16	6	0.8%	0.00	1.00
ENZL	cluster_1	Developed ex-US Equity	2026-07-07	13	2.5%	0.00	1.00
VYMI	cluster_1	Developed ex-US Equity	2026-07-08	12	2.1%	0.00	1.00
EWZ	cluster_1	Developed ex-US Equity	2026-07-16	6	1.1%	0.00	1.00
IDLV	cluster_1	Developed ex-US Equity	2026-07-10	10	0.9%	0.00	1.00
DWX	cluster_1	Developed ex-US Equity	2026-07-09	11	1.1%	0.00	1.00
IPKW	cluster_1	Developed ex-US Equity	2026-07-10	10	1.3%	0.00	1.00
EFV	cluster_1	Developed ex-US Equity	2026-07-08	12	1.8%	0.00	1.00
PID	cluster_1	Developed ex-US Equity	2026-07-16	6	-0.5%	0.00	1.00
EWS	cluster_1	Developed ex-US Equity	2026-06-17	25	7.0%	0.00	1.00
EWC	cluster_1	Developed ex-US Equity	2026-07-10	10	0.7%	0.01	0.99
RWX	cluster_1	Developed ex-US Equity	2026-07-16	6	0.2%	0.01	0.99
EWA	cluster_1	Developed ex-US Equity	2026-07-20	4	0.7%	0.00	0.99
NFRA	cluster_1	Developed ex-US Equity	2026-07-16	6	0.5%	0.00	0.96
EWU	cluster_1	Developed ex-US Equity	2026-07-16	6	0.6%	0.04	0.95
EUFN	cluster_1	Developed ex-US Equity	2026-06-12	28	3.9%	0.10	0.90
VNQI	cluster_1	Developed ex-US Equity	2026-07-20	4	-0.4%	0.22	0.77
QEFA	cluster_1	Developed ex-US Equity	2026-07-06	14	-0.8%	0.23	0.74
IGF	cluster_1	Developed ex-US Equity	2026-07-09	11	1.0%	0.25	0.73
VIGI	cluster_1	Developed ex-US Equity	2026-06-30	17	1.9%	0.34	0.63
RODM	cluster_1	Developed ex-US Equity	2026-07-20	4	0.7%	0.02	0.63
IQLT	cluster_1	Developed ex-US Equity	2026-07-01	16	0.9%	0.39	0.50
IVLU	cluster_1	Developed ex-US Equity	2026-07-15	7	-0.4%	0.08	0.49
INTF	cluster_1	Developed ex-US Equity	2026-07-15	7	-0.8%	0.22	0.42
GSIE	cluster_1	Developed ex-US Equity	2026-07-02	15	-0.0%	0.37	0.42
VGK	cluster_1	Developed ex-US Equity	2026-07-01	16	0.7%	0.47	0.09
IEUR	cluster_1	Developed ex-US Equity	2026-07-15	7	-0.9%	0.53	-0.11
IEV	cluster_1	Developed ex-US Equity	2026-07-01	16	0.5%	0.64	-0.14
EWO	cluster_1	Developed ex-US Equity	2026-04-10	72	8.7%	0.91	-0.16
IQDF	cluster_1	Developed ex-US Equity	2026-07-15	7	-1.6%	0.31	-0.18
ESGD	cluster_1	Developed ex-US Equity	2026-06-30	17	-0.2%	0.66	-0.27

EFA	cluster_1	Developed ex-US Equity	2026-07-01	16	0.4%	0.69	-0.47
IEFA	cluster_1	Developed ex-US Equity	2026-07-15	7	-1.2%	0.67	-0.53
AMLX	cluster_12	Materials	2026-07-10	10	4.3%	0.00	1.00
MLPA	cluster_12	Materials	2026-07-10	10	4.2%	0.00	1.00
MOO	cluster_12	Materials	2026-07-07	13	2.8%	0.00	1.00
IGE	cluster_12	Materials	2026-07-23	1	0.3%	0.00	1.00
MLPX	cluster_12	Materials	2026-07-09	11	2.7%	0.00	1.00
EMLP	cluster_12	Materials	2026-07-08	12	1.7%	0.00	1.00
NANR	cluster_12	Materials	2026-07-24	0	0.0%	0.00	0.99
PPH	cluster_11	Healthcare	2026-05-21	43	6.6%	0.56	0.44
XLV	cluster_11	Healthcare	2026-05-21	43	9.7%	0.62	0.38
IYH	cluster_11	Healthcare	2026-05-20	44	10.5%	0.71	0.29
FHLC	cluster_11	Healthcare	2026-05-19	45	11.3%	0.71	0.29
VHT	cluster_11	Healthcare	2026-05-19	45	11.3%	0.74	0.26
IHF	cluster_11	Healthcare	2026-05-08	52	13.0%	0.75	0.25
PTH	cluster_11	Healthcare	2026-06-17	25	10.9%	0.92	0.08
PSCH	cluster_11	Healthcare	2026-04-13	71	23.5%	0.93	0.07
XMLV	cluster_2	U.S. Mid Cap Equity	2026-06-25	20	4.5%	0.00	1.00
REGL	cluster_2	U.S. Mid Cap Equity	2026-06-12	28	4.0%	0.10	0.90
SMMV	cluster_2	U.S. Mid Cap Equity	2026-06-25	20	3.5%	0.35	0.65
IJS	cluster_2	U.S. Mid Cap Equity	2026-04-10	72	10.6%	0.37	0.63
XRT	cluster_2	U.S. Mid Cap Equity	2026-05-28	39	1.6%	0.71	0.28
SDVY	cluster_2	U.S. Mid Cap Equity	2026-06-08	32	4.6%	0.76	0.24
ISMD	cluster_2	U.S. Mid Cap Equity	2026-04-09	73	15.9%	0.84	0.16
FTXR	cluster_2	U.S. Mid Cap Equity	2026-04-17	67	8.8%	0.76	0.14
RTH	cluster_2	U.S. Mid Cap Equity	2026-07-17	5	-2.1%	0.65	-0.31
DTD	cluster_3	U.S. Large Cap Value	2026-04-13	71	7.1%	0.00	1.00
DLN	cluster_3	U.S. Large Cap Value	2026-04-13	71	7.0%	0.01	0.99
FNDX	cluster_3	U.S. Large Cap Value	2026-04-09	73	9.4%	0.02	0.98
DGRO	cluster_3	U.S. Large Cap Value	2026-04-13	71	7.7%	0.13	0.87
RWL	cluster_3	U.S. Large Cap Value	2026-04-10	72	10.0%	0.14	0.86
IWX	cluster_3	U.S. Large Cap Value	2026-04-10	72	12.7%	0.15	0.85
PKW	cluster_3	U.S. Large Cap Value	2026-05-20	44	6.2%	0.16	0.84
VONV	cluster_3	U.S. Large Cap Value	2026-04-10	72	11.8%	0.24	0.76
IWD	cluster_3	U.S. Large Cap Value	2026-04-10	72	11.9%	0.26	0.74
PRF	cluster_3	U.S. Large Cap Value	2026-04-09	73	10.0%	0.28	0.72
VTV	cluster_3	U.S. Large Cap Value	2026-04-13	71	9.1%	0.62	0.38
VIG	cluster_3	U.S. Large Cap Value	2026-04-14	70	6.6%	0.69	0.29
AIVL	cluster_3	U.S. Large Cap Value	2026-04-10	72	9.6%	0.75	0.25
RSP	cluster_3	U.S. Large Cap Value	2026-04-14	70	7.0%	0.85	0.14
RSPM	cluster_3	U.S. Large Cap Value	2026-06-16	26	-2.0%	0.58	-0.45
ITA	cluster_3	U.S. Large Cap Value	2026-07-24	0	0.0%	0.98	-0.70
FIDU	cluster_3	U.S. Large Cap Value	2026-07-23	1	0.1%	0.99	-0.78
VIS	cluster_3	U.S. Large Cap Value	2026-07-23	1	0.1%	1.00	-0.80
FEX	cluster_3	U.S. Large Cap Value	2026-04-09	73	7.5%	0.99	-0.95

EWI	cluster_13	China Equity	2026-07-23	1	0.5%	0.00	1.00
KURE	cluster_13	China Equity	2026-07-06	14	0.4%	0.01	0.99
SIVR	cluster_17	Gold	2026-05-27	40	-22.1%	0.21	-0.21
SLV	cluster_17	Gold	2026-05-27	40	-22.1%	0.21	-0.21
BIL	cluster_24	Cash / Ultra-Short Fixe...	2026-07-16	6	0.1%	0.18	0.59
SHV	cluster_24	Cash / Ultra-Short Fixe...	2026-07-20	4	0.0%	0.18	0.53
GBIL	cluster_24	Cash / Ultra-Short Fixe...	2026-07-20	4	0.0%	0.20	0.41
MINT	cluster_24	Cash / Ultra-Short Fixe...	2026-07-20	4	0.0%	0.32	0.27
ICSH	cluster_24	Cash / Ultra-Short Fixe...	2026-07-21	3	-0.0%	0.33	0.10
FTSM	cluster_24	Cash / Ultra-Short Fixe...	2026-07-21	3	0.0%	0.30	0.05
IOO	cluster_5	U.S. Large Cap Equity	2026-07-15	7	-2.3%	0.18	0.06
OEF	cluster_5	U.S. Large Cap Equity	2026-07-06	14	-1.7%	0.51	-0.32
URTH	cluster_5	U.S. Large Cap Equity	2026-07-01	16	-0.6%	0.65	-0.41
TUR	cluster_20	India Equity	2026-07-24	0	0.0%	0.24	0.56
FXB	cluster_25	International Fixed Inc...	2026-07-15	7	-1.6%	0.17	0.76
EWJV	cluster_22	Japan Equity	2026-07-06	14	0.2%	0.06	0.88
HEWJ	cluster_22	Japan Equity	2026-07-24	0	0.0%	0.97	-0.84
DBJP	cluster_22	Japan Equity	2026-07-24	0	0.0%	0.95	-0.85
SDIV	cluster_8	Emerging Markets Equity	2026-07-21	3	-0.6%	0.00	1.00
DVYE	cluster_8	Emerging Markets Equity	2026-07-24	0	0.0%	0.00	1.00
FNDE	cluster_8	Emerging Markets Equity	2026-07-23	1	-0.3%	0.01	0.76
AIA	cluster_8	Emerging Markets Equity	2026-07-10	10	-5.5%	0.92	-0.92
SRLN	cluster_23	Bank Loan / Floating Ra...	2026-07-22	2	-0.1%	0.01	0.70
FLRN	cluster_23	Bank Loan / Floating Ra...	2026-07-13	9	0.1%	0.29	0.31
FLOT	cluster_23	Bank Loan / Floating Ra...	2026-07-10	10	0.1%	0.31	0.20
PTF	cluster_6	U.S. Large Cap Growth	2026-04-07	75	10.0%	0.98	-0.98
QTUM	cluster_6	U.S. Large Cap Growth	2026-04-10	72	19.6%	1.00	-1.00
IBDR	cluster_4	Aggregate Fixed Income	2026-07-21	3	0.0%	0.22	0.32

ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

Top 20 most bullish (cluster-deduped, abs-mom approved)

ticker	cluster	theme	p_buy	nss
AML	cluster_12	Materials	1.00	1.00
FTGC	cluster_19	Broad Commodities	1.00	1.00
RDIV	cluster_9	U.S. Dividend Equity	1.00	1.00
RFV	cluster_2	U.S. Mid Cap Equity	1.00	1.00
EPP	cluster_1	Developed ex-US Equity	1.00	1.00
RWO	cluster_15	Real Estate	1.00	1.00
VOE	cluster_3	U.S. Large Cap Value	1.00	1.00
SDIV	cluster_8	Emerging Markets Equity	1.00	1.00
PBJ	cluster_18	Consumer Staples	1.00	0.92
EWJV	cluster_22	Japan Equity	0.94	0.88
KCE	cluster_14	Financials	0.98	0.84
QDF	cluster_5	U.S. Large Cap Equity	1.00	0.82
FLTR	cluster_23	Bank Loan / Floating Rate	0.94	0.79
IPAY	cluster_7	U.S. Extended Market Equi...	1.00	0.76
BIL	cluster_24	Cash / Ultra-Short Fixed ...	0.76	0.59
PPH	cluster_11	Healthcare	0.99	0.44
IBDR	cluster_4	Aggregate Fixed Income	0.54	0.32
SGOL	cluster_17	Gold	0.00	-0.10
SCHG	cluster_6	U.S. Large Cap Growth	0.31	-0.34
STIP	cluster_21	Inflation-Protected Fixed...	0.00	-0.64

Top 20 most bearish (cluster-deduped)

ticker	cluster	theme	p_sell	nss
PCY	cluster_10	High Yield Bond	1.00	-1.00
ITM	cluster_16	Municipal Bond	1.00	-1.00
VNM	cluster_20	India Equity	1.00	-1.00
EWX	cluster_8	Emerging Markets Equity	1.00	-1.00
GTO	cluster_4	Aggregate Fixed Income	1.00	-1.00
CNXT	cluster_13	China Equity	1.00	-1.00

VTWG	cluster_7	U.S. Extended Market Equi...	1.00	-1.00
IGBH	cluster_23	Bank Loan / Floating Rate	1.00	-1.00
QTUM	cluster_6	U.S. Large Cap Growth	1.00	-1.00
AOK	cluster_5	U.S. Large Cap Equity	1.00	-1.00
LTPZ	cluster_21	Inflation-Protected Fixed...	1.00	-1.00
EFG	cluster_1	Developed ex-US Equity	1.00	-1.00
IYC	cluster_2	U.S. Mid Cap Equity	1.00	-1.00
FXV	cluster_25	International Fixed Income	1.00	-1.00
PKB	cluster_3	U.S. Large Cap Value	1.00	-1.00
UNG	cluster_19	Broad Commodities	1.00	-1.00
VPL	cluster_22	Japan Equity	1.00	-1.00
GSY	cluster_24	Cash / Ultra-Short Fixed ...	0.99	-0.99
URA	cluster_12	Materials	0.98	-0.98
SILJ	cluster_17	Gold	0.61	-0.61

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_20	10	FLAT	0.305	0.095	0.00022	
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_19	11	FLAT	0.480	0.076	0.00017	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.741	0.038	0.00008	
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_25	6	FLAT	0.729	0.095	0.00022	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.634	0.213	0.00036	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_19	11	LONG	0.598	0.168	0.00039	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.743	0.126	0.00031	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.743	0.153	0.00040	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.699	0.191	0.00045	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.679	0.179	0.00046	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.714	0.177	0.00044	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_20	10	LONG	0.500	0.218	0.00056	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.869	0.020	0.00001	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_4	58	FLAT	0.847	0.127	0.00036	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.869	0.075	0.00021	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_4	58	FLAT	0.876	0.161	0.00031	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_25	6	FLAT	0.820	0.205	0.00049	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_4	58	FLAT	0.832	0.233	0.00047	
NOBL	ProShares S&P 500 Dividend Aris...	U.S. Dividend Equity	cluster_20	10	FLAT	0.447	0.273	0.00075	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_16	13	FLAT	0.877	0.248	0.00046	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_20	10	FLAT	0.447	0.327	0.00083	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_13	15	LONG	0.791	0.269	0.00071	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_16	13	FLAT	0.877	0.263	0.00054	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_12	17	FLAT	0.746	0.360	0.00085	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_20	10	FLAT	0.494	0.348	0.00102	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_23	9	FLAT	0.865	0.323	0.00071	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_20	10	FLAT	0.557	0.405	0.00107	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_13	15	FLAT	0.857	0.256	0.00073	
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected F...	cluster_21	10	FLAT	0.918	0.226	0.00057	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_13	15	FLAT	0.904	0.238	0.00065	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
imminent_horizon	screen	imminent_horizon	1
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20